ENVS 193DS Final

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Link to Github repo

https://github.com/hparsons333/ENVS-193DS_spring-2026_final

Setting up

```r
library(tidyverse)
library(here)
```

Problem 1. Research writing

You're working on a research team trying to understand how restoration of native plant cover (measured in percent cover) influences probability of American Avocet (Recurvirostra americana) and other shorebird habitat use in the San Francisco Bay. Your co-worker runs some analyses and writes up a report, giving it to you to review. In part 1 of the results section of the report, your co-worker has written:

We rejected the null hypothesis that distance to water edge does not predict probability of American Avocet microhabitat use (p = 0.03).

In part 2 of the results section of the report, your co-worker has written:

We rejected the null hypothesis that there is no relationship between bird species (American Avocet, Forster's Tern, Black-necked Stilt) and habitat (managed pond, non-tidal marsh, salt pond, tidal marsh, or tidal mudflat) (p < 0.001).

It's great that this report is coming together, but you think they can improve on what they've written and make it more understandable to a non-statistical audience.

1a. Transparent statistical methods

What statistical tests did your co-worker use?

Clearly connect the test to the part that you are addressing (e.g. "In part 1, they used ___________. In part 2, they used ___________.").

In part 1, they used a simple linear regression to test if distance to water edge predicted American Avocet microhabitat use. In part 2, they used a correlation coefficient test to see if there was a relationship between bird species and habitat.

1b. Suggestions for rewriting

In 1-3 sentences, write new statements with interpretation only for the statements in part a and b.

Be sure your interpretation makes sense without any statistical summary (which, again, you should not include).

Rewriting statement a

Our results suggest that we can reject the null hypothesis that distance to water edge does not predict probability of American Avocet microhabitat use. Alternatively, we determine that the distance to water edge significantly predicts whether or not American Avocent will utilize microhabitats.

Rewriting statement b.

Our results suggest that we can reject the null hypothesis, which states that there is no relationship between the type of bird species found and the habitat type. Alternatively, we determine that there is a significant relationship between the type of bird species found and the habitat type.

1c. Further analysis

In 1-2 sentences, outline:

- one additional variable that could influence the probability of American Avocet microhabitat use

- what type of variable it is (with units and/or levels, if applicable)

- why that variable might influence American Avocet habitat use.

Be specific with units, data collection method, etc. as needed.

Because American Avocet need their nesting, brooding-rearing, foraging, and roosting sites to be protected from predators, another variable that could influence the probability of American Avocet habitat use is island distance from landmass/shore, since increased distance from the nearest landmass would provide increased distance from larger predators.

This would be a continuous variable, measured in distance in kilometers (since the area of the pond is measured in kilometers squared).

Problem 2. Data analysis

Reading in data

```
nests <- read_csv(here("data", "nest_data_final.csv"))
```